

Impact of Feature Scaling and Imputation on k -Nearest Neighbor Performance in Retail Analytics

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I. INTRODUCTION

The primary objective of this paper is to develop a prediction model based on a machine learning algorithm, specifically k -Nearest Neighbors (kNN) to classify customers of their probability or likelihood of purchasing a product through proper utilization of a customer dataset from four separate store regions. The model is designed to analyze the dataset and separate the customers into two groups which are customers who would likely to purchase, and customers who are not. Through the analyzation of the dataset, which includes their demographic information, transaction history, and their behavioral data. In addition, predicting their likelihood of purchase can be beneficial to aid on making data-driven decisions, such as targeted marketing and resource allocation.

To ensure accurate prediction and reliability from the kNN model, as its reputation of being highly sensitive to noise, and outliers, and scale differences [1], the dataset undergoes preparation and cleaning. Through observation, the raw data from four the store regions contain inconsistencies, missing values, redundant records, and logical errors.

A. Inconsistent Categorical Labels

The dataset presents inconsistencies specifically, on the *Gender* column, as it contains significant fragmentation such as 'Male', 'm', 'M', 'Female', 'F', and 'Fem' which can dilute predictive-signals as it will treat it as six distinct genders rather than two [2], hence correcting process is required. The inconsistencies go the same with the columns such as *Store_Region* and *Device_Type*, containing capitalization errors and additional whitespace as well as the date format from *Purchase_Dates*.

B. Redundant Records

The dataset also presents redundant records, to be specific there 14 records under the *Customer_ID* column appears twice specifically, C-1019, C-1039, C-1059, C-1079, C-1099, C-1119, C-1139, C-1159, C-1179, C-1199, C-1219, C-1239, C-1259, and C-1279 which can cause conflict as there the records also present inconsistency and contradiction on their values like *Age*, *Gender*, and *Income* in which can confuse the algorithm.

C. Invalid Data and Outliers

Outliers are hindrance to the kNN model as they skew the normalization process, as they are columns on the dataset that contains negative as well as impossible ages which are

Age, *Annual_Income* and *Membership_Score* as such values appears frequently, which pertains to systematic errors and logically impossible which requires to be corrected.

D. Missing Values

The dataset presents challenges in missingness in two of the columns namely, the *Age* and *Total_Spend*. In *Total_Spend*, the column is missing 70.6% records out of 300 while, *Age* contains 70 missing entries. This missingness can result in inconsistencies as kNN will have difficulties in computing distances if features contain null values.

E. Scale Imbalance

A big difference between the range of the values for both *Annual_Income* and *Support_Calls* can cause bias as the income is varying from about -4,825 to 121,577, in contrast to the support calls which can be between 0 and 10. This may present an issue for the kNN algorithm as it uses Euclidean distance as they large *Annual_Income* values may overpower the *Support_Calls*.

II. METHODOLOGY

A. Preparation Pipeline

The dataset preparation process involves a modular pipeline broken into seven distinct tasks designed to transform the raw, noisy dataset into clean, and numerical format in order to aid the kNN algorithm

1) *Data Loading and Initial Cleanup*: The script starts by loading the dataset from the csv file *adjusted_noisy_knn_300.csv* and importing the necessary libraries (*pandas*, *numpy*, and *sklearn*) proceed with ensuring that there are no duplicate entries or records *drop_duplicates* that can cause bias to the model and ensure that the training set represents a distinct customer.

2) *Handling Outliers and Missing Values*: The code proceeds with indentifying unrealistic values such as negative ages or ages over 100 and replaces them with *NaN*. In addition all missing values such as *Age*, *Annual_Income*, and *Total_Spend* are imputed using the median in which preferred over the mean since it is not susceptible to outliers preventing extreme values from distorting to the dataset, subsequently preserving valuable data points that otherwise would be lost.

3) *String Standardization*: Columns such as *Gender*, *Device_Type*, will undergo normalization process to fix

inconsistencies caused by human errors. Hence, the code strips the extra whitespace, converts text to lowercase, and maps various labels such as 'Male', 'm', 'M', 'Female', 'F', and 'Fem' to standard categories. This can ensure that the dimensionality of the data will be reduced through ensuring that the model will be able to properly identify that 'mobile' and 'Mobile' are the same.

4) *Binning and Merging*: The dataset undergoes transforming and is binned into categorical groups (*low*, *medium*, and *high*) to aid the model to detect non-linear behavioral patterns. In addition, external *Tax_Rate* is merged based on the *Store_Region* to add relevant context in regards to economy to the model.

5) *Normalization*: MinMaxScaler class is applied to all numerical values such as *Age*, and *Annual_Income*, and transform them into a standardized value ranging from 0 to 1. This can aid the kNN algorithm to avoid biases on larger values and would avoid overpowering the smaller value such as *Support_Calls*. In addition, this is one of the most critical steps for the kNN algorithm as it heavily relies on Euclidean distance to find neighbors and the transformation is performed utilizing this formula below.

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

6) *Categorical Encoding*: Machine learning models require numerical input, hence all categorical variables such as *Store_Region*, *Gender*, and the created *Purchase_Volume* which is based on *Item_Purchased* are converted into binary columns using *pd.get_dummies* in which translates qualitative descriptions into a quantitative format of 0s and 1s that the algorithm can process mathematically.

7) *Final Cleanup and Export*: This process removes the list of columns that are no longer needed for the model which are 'Membership_Score', 'Newsletter_Sub', 'Customer_ID', 'Purchase_Date' as it only carries little to no behavioral patterns useful for classification and will only serve as additional noise. After which, the fully processed dataset is then saved as *cleaned_knn_data.csv*

B. Experimental Setup

The experiment setup compares two modeling approaches to assess the impact of data preparation on predictive performance. The Baseline Run acts as a control, naively training a kNN classifier on the raw dataset by selecting only numeric columns and dropping any rows with missing values, thereby discarding valuable categorical predictors. In contrast, the Optimized Run utilizes the fully prepared dataset where missing values are imputed, categorical variables are one-hot encoded, and all features are normalized to a standard 0-1 scale to ensure equal weighting. To evaluate model stability and determine the optimal neighborhood size, both runs were tested across a range of odd k-values from 1 to 20. An identical 80/20 train-

test split was maintained throughout to isolate data quality as the primary variable driving performance differences.

III. RESULTS AND ANALYSIS

Through analysis of the dataset before and after the preparation pipeline, it reveals significant improvements in data integrity, which directly correlate with the learning ability of the model.

A. Data Quality Comparison

- Before Preparation, the dataset is heavily compromised by the noise, as it contained 282 missing values primarily in *Age* and *Total_Spend*, and 14 duplicate customer records, which definitely introduced bias. In addition, the *Age* column indicates data entry errors such as -10 to 200 which clearly indicates invalid data, as well as the *Gender* containing multiple labels 'Male', 'm', 'M', 'Female', 'F', and 'Fem' which only pertains to two categories
- After Preparation, the pipeline successfully standardized the dataset. All duplicate records were removed, reducing the count to 286 unique customer observations. Missing values were handled via median imputation, ensuring zero data loss for valid records. Outliers were capped, and inconsistent text labels were unified and converted into 21 numerical features (via one-hot encoding). This transformation produced a dense, mathematical-ready matrix with features scaled between 0 and 1.
- Further optimizing the dataset, through strict outlier handling such as dropping the entire row instead of relying on best guess through median, cleaning the whitespaces on the *Store_Region* column refining the binning ensuring that it is equally distributed from [0, 15, 35, 51] to [0, 17, 34, 51], removing *Total_Spend* and *Support_Calls* and keeping the *Membership_Score* and *Promo_User*, a better imputation logic, adding Boolean conversion to *Promo_User*, will ensure a higher quality dataset reducing the records to 66

B. Performance Comparison

In order to compare the impact of the data preparation pipeline, a comparison of the predictive performance of the Baseline model against the Optimized model is required and both models utilized a fixed hyperparameter of $k = 5$ in order to ensure that a direct comparison in regards to data quality.

Metric	Baseline Run (Raw)	Optimized Run (Prepared)
Accuracy	42.86%	68.97%
Confusion Matrix	[[3, 1], [7, 3]]	[[22, 9], [9, 18]]
True Positives (Buy)	3	18
True Negatives (No Buy)	3	22
False Negatives (Missed)	7	9
Observations Used	14 (Deleted NaNs)	58 (Imputed)

Figure 1.

The result as stated on Figure 1., demonstrates a dramatic improvement in the model reliability. The Baseline model yielded an accuracy of only 42.86%, which is poor

performance is a result of the deletion strategy for missing values, which reduced the test set to only 14 observations, and the lack of feature scaling, which allowed high-magnitude variables like *Annual_Income* to distort the distance calculations.

In contrast, the Optimized model achieved an accuracy of 68.97%, an improvement of over 26 percentage points. The Confusion Matrix reveals a balanced classification capability, with the model correctly identifying 22 non-purchasers (True Negatives) and 18 purchasers (True Positives). This confirms that imputation and normalization process done which considered the critical steps for enabling the kNN algorithm to interpret complex behavioral patterns effectively.

C. K-Optimization

In order to determine the optimal k for maximizing the prediction accuracy the optimized model was tested across all odd values of k from 1 to 20.

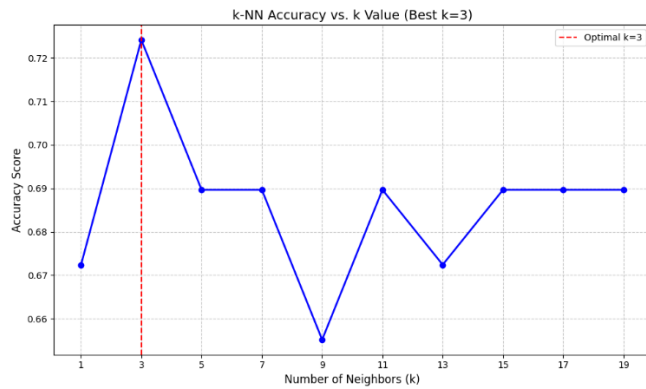


Figure 2.

As stated on the Figure 2., the analysis of k values identifies that setting the value as 3 is the optimal hyperparameter for this dataset that resulted in achieving a peak accuracy of 72.41%. In addition, increasing the k value beyond 5, can result to 68.97% although, dips down to 65.52% when choosing 9, then steadily ramps up back to 68.97% on 11, while on 13 the performance dips to 67.24% then steadily went up to 68.97% as it reaches 19. This indicates that the model performance is dependent on a small neighborhood size, which indicates the customer that the customer purchasing behavior is highly localized.

--- Optimized k-NN Model Results ---
Accuracy Score: 85.71%

Confusion Matrix:
[[4 0]
[2 8]]

Classification Report:

	precision	recall	f1-score	support
0	0.67	1.00	0.80	4
1	1.00	0.80	0.89	10
accuracy			0.86	14
macro avg	0.83	0.90	0.84	14
weighted avg	0.90	0.86	0.86	14

Figure 3.

Through further optimization, as stated on the Figure 3., the accuracy score yielded at 85.71% utilizing k value as 5, although this resulted into a great result having a dataset record reduced to 66, maybe considered non-feasible as it lacks of statistical power or the small number of observations.

IV. RESULTS AND ANALYSIS

A. Reflections

Based on the experiment conducted, determining the noise by face-value alone can yield into a higher accuracy, although through rigorous analysis and observing the customer records, a highly fragmented values were faced like in the *Gender* column. In addition, the whitespaces are not obvious at first that which are text inconsistencies that can lead to the algorithm to treat it as a separate category. Moreover, understanding how the exact code for the data preparation pipeline, and the Implementation kNN algorithm on both the raw and optimized dataset the first time is a rigorous process. Additionally, choosing k value is a difficult yet enjoying task allow you to feel satisfaction as you go along with the exercise. Lastly, through observation, noticing that the bins are not equally distributed is also a factor that can validate the knowledge learned from the previous discussions.

B. Final Takeaway

The experiment signifies that Data Quality drives the model success. As the experiment proves that flawed data made the “Baseline” model worse, while on the other-hand proper preprocessing of the dataset can yield to a good result, hence consistency on how the data is gathered and ensuring that it is consistent on every part of the business can produce a model successful enough on making business data-driven decision making.

REFERENCES

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